



## Still Lit, Still Pressed: A Pre/Post Habituation Study of Door-Close Delay Retuning and Redundant Call-Button Presses

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**Abstract.** Facilities retuned the door-close delay on the hall call buttons across the tallest building's lift lobbies in Semester 1 2026, shortening the delay from 4.5 to 2.2 seconds to bring the building's floor-throughput figure back within the Living Dashboard's published target band. We ask whether the shorter delay changed how often riders re-pressed an already-lit call button — logged by the dispatch controller but never reported alongside the throughput figure it accompanies — and whether any change faded as riders learned the new timing. Using a quasi-experimental pre/post design over four baseline weeks and twelve post-change weeks (39,092 logged hall calls), we find the retuned delay produced an immediate 2.3-fold rise in the redundant-press rate, which then declines over the following weeks toward a fitted asymptote that remains 2.8 percentage points above baseline (half-life 4.9 weeks). The decay half-life does not differ reliably across low, mid, and high floor bands. The latency between an initiating press and the first redundant press compresses over the same window. The floor-throughput figure that motivated the retuning has sat within its target band throughout the post-change period we logged; no tile in the Living Dashboard's own suite carries the redundant-press rate, or anything else describing what riders made of the change.

### Introduction

Lift dispatch controllers expose a small number of tunable parameters to the buildings that operate them, and door-close delay — the interval a car holds its doors open at a floor before beginning to close — is among the more consequential. A shorter delay serves more floor-stops per hour; a longer one gives riders more time to board, exit, or simply arrive. Facilities Division treats the parameter as a lever on a single published figure: the floor-throughput tile on the Living Dashboard, tracked against a target of at least 46 completed floor-stops per lift car per peak hour across the tallest building's four cars.

In Semester 1 2026, following a run of weeks in which the tallest building's floor-throughput figure sat below its target band, Facilities retuned the door-close delay down from 4.5 to 2.2 seconds. The dispatch controller's own hall-call

log — part of the same telemetry stream audited in prior institutional work on this building's car-level energy-saving claims [1] — also records every press against every hall call button, including presses registered while that direction's call is already lit. This paper asks a question the throughput figure was never built to answer: did the shorter delay change how often riders pressed an already-lit button, and if it did, did that change persist as riders learned the new timing, or fade back toward whatever counted as normal before the retuning?

We treat a redundant press — any press on a hall call button already active for more than one second — as a behavioural signal distinct from the throughput figure the retuning targeted, and track it across a four-week pre-change baseline and twelve weeks following the change. The redundant press is a poor proxy for anything Facilities has asked of it; it is, we suggest, a rea-

sonably direct proxy for whether a rider believed their first press had registered.

We make three contributions, each hedged appropriately:

- We report a quasi-experimental estimate of the immediate effect of a door-close delay change on redundant call-button presses, using segmented pre/post logging rather than a single before/after comparison.
- We fit a habituation decay curve to the twelve post-change weeks and estimate both a half-life and an asymptote, finding the redundant-press rate settles above, not at, its pre-change baseline.
- We show the decay is consistent across floor bands, suggesting the effect reflects the delay change itself rather than any one part of the building.

The remainder of the paper reviews related work on dispatch tuning, perceived control, and habituation; describes the logging and decay-model method; reports the pre/post and decay results; and discusses what a redundant press can and cannot tell an institution that only ever asked its dashboard.

## Related work

Elevator dispatch research has long optimised for throughput and waiting-time objectives, from early reinforcement-learning formulations [2] through Q-learning and recurrent architectures tuned to building-specific traffic patterns [3], to recently deployed systems reporting performance gains against rule-based baselines [4]. Across this literature, the passenger is a source of arrival-time load on the objective function; a controller is judged on the figure it moves, not on whether the humans generating that figure changed how they behaved around it.

Riders press call buttons in part because pressing confers a sense of agency over an outcome the mechanism, not the press, actually determines — a pattern consistent with the illusion-of-control literature [5], and with field evidence that restoring even small increments of perceived control changes measurable behaviour [6]. Confirmation feedback on a control input has itself been shown to change how a population uses that input: adding audible and visual confirmation to pedestrian crossing buttons increased both button use and compliance with the resulting signal [7], motivating our own inter-

est in what happens when a control input's feedback timing changes instead. Prior institutional work examined an analogous instrumented control input directly against ground truth, finding a pedestrian crossing button's press counts bore no reliable association with the signal controller's own logged phase changes [8]; the present study takes the same approach from the dispatch side, treating the press log itself, rather than a count derived from it, as the object of study. Human-computer interaction has long treated a roughly two-second boundary as the point past which a lack of feedback reads as unresponsiveness rather than delay [9].

The behavioural decay we model draws on the dual-process account of habituation, in which a response to a repeated, non-threatening stimulus change is modelled as the net outcome of competing habituation and sensitisation processes rather than a single decay term [10], refined in a later consensus description of habituation's behavioural characteristics [11]. Building-performance research has begun assembling occupant-centric key-performance-indicator frameworks explicitly distinct from purely operational metrics such as throughput [12], and the corrupting pressure a measure comes under once treated as a target, articulated originally by Campbell [13] and demonstrated empirically in academic-metric settings [14], motivates our choice to log a behaviour the throughput retuning never asked about.

## Method

All hall call buttons across the tallest building's lift lobbies are wired through a single dispatch controller that timestamps every press, every door-open and door-close event, and every car assignment. We define a redundant press as any press registered against a hall call button already active (illuminated) for that direction, more than one second after the call's initiating press — the one-second guard removes near-simultaneous double-contact presses that reflect a mechanical bounce rather than a rider's second decision to press.

The controller retuned its door-close delay from 4.5 to 2.2 seconds on the first Monday of Semester 1 2026, as part of a scheduled parameter review triggered by the floor-throughput tile sitting below its 46-floor-stops-per-car-per-peak-hour target for three consecutive weeks. We

extracted the full hall-call log for a four-week pre-change baseline and the twelve weeks immediately following the change, aggregating redundant presses into weekly bins against each week's total logged calls to produce a weekly redundant-press rate per 100 calls. Our four-week baseline and twelve-week post-change window follows the segmented pre/post logic of interrupted time-series designs [15], treating the delay change as a single well-dated intervention rather than inferring an effect from a single before/after mean.

We fit the twelve post-change weekly rates to an exponential decay model,

$$R(t) = R_{\infty} + (R_0 - R_{\infty})e^{-t/\tau}$$

where  $R(t)$  is the redundant-press rate in post-change week  $t$ ,  $R_0$  the fitted immediate post-change rate,  $R_{\infty}$  the fitted long-run asymptote, and  $\tau$  a time constant from which we report the half-life  $\tau \ln 2$ . Parameters were estimated by nonlinear least squares against the twelve weekly means, with the four-week baseline mean held out as an independent check on whether  $R_{\infty}$  recovers it.

To test whether the decay is a property of the delay change itself rather than of any one part of the building, we repeated the fit separately for calls originating on low floors (1–4), mid floors (5–8), and high floors (9–12), comparing the three fitted half-lives against their bootstrap confidence intervals (2,000 resamples per band). We additionally logged, for every redundant press, the latency between the initiating press and the first redundant press, and compare the pre- and post-change latency distributions directly rather than through the decay model, since latency is defined only conditional on a redundant press having already occurred.

## Results

The four-week baseline logged 9,844 hall calls and 827 redundant presses, a stable rate of 8.4 per 100 calls (weekly range 8.1–8.7). In the first week following the retuning, the redundant-press rate rose to 19.6 per 100 calls — a 2.3-fold increase — before declining across the following eleven weeks (Figure 1). The fitted decay model estimates  $R_0 = 19.4$ ,  $R_{\infty} = 11.2$ , and  $\tau = 7.1$  weeks, giving a half-life of 4.9 weeks; the fitted asymptote sits 2.8 percentage points

above the four-week baseline mean held out of the fit, indicating the population does not fully re-learn the new timing within the twelve weeks observed.

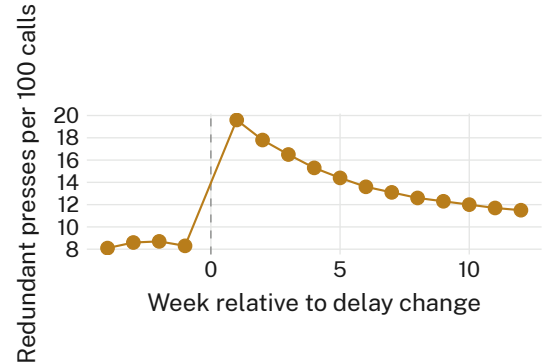


Figure 1: Weekly redundant-press rate, four baseline weeks against twelve weeks following the door-close delay retuning (dashed line); the rate more than doubles immediately, then decays with an estimated 4.9-week half-life toward an asymptote above baseline.

| Period           | Weeks | Hall calls | Redundant presses | Rate/100 |
|------------------|-------|------------|-------------------|----------|
| Baseline         | 4     | 9,844      | 827               | 8.4      |
| Post, week 1     | 1     | 2,690      | 527               | 19.6     |
| Post, weeks 2-12 | 11    | 26,558     | 3,051             | 11.5     |

Table 1: Logged hall calls and redundant presses by period.

Splitting the decay fit by the floor band a call originated on shows overlapping half-lives — 4.7 weeks (low floors), 5.1 weeks (mid floors), and 4.8 weeks (high floors) — with bootstrap confidence intervals that overlap pairwise across all three bands (Figure 2). We read this as evidence that the decay is a property of the delay change rather than of any one part of the building's usage pattern; we retain the floor-band split for completeness.

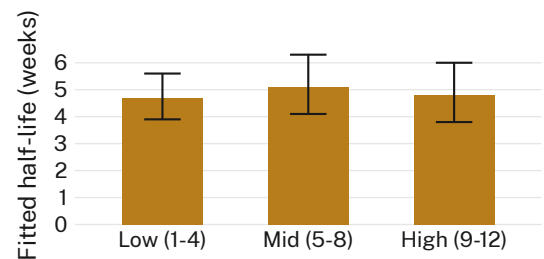


Figure 2: Fitted habituation half-life by call-origin floor band, with bootstrap 95% confidence intervals; the three bands are statistically indistinguishable.

The latency between an initiating press and the first redundant press compresses alongside the shorter delay: a pre-change median of 2.8 seconds (IQR 1.9–3.6) falls to a post-change median of 1.6 seconds (IQR 1.1–2.3) (Figure 3), consistent with riders re-pressing sooner once the window in which a first press could plausibly still be “working” itself shortened.

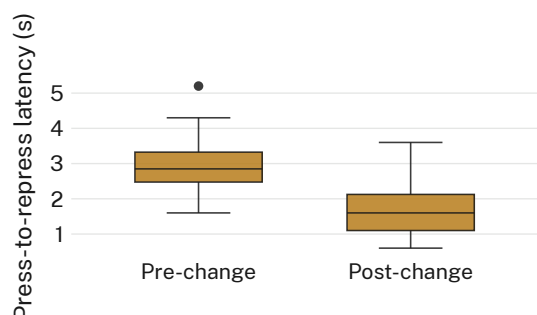


Figure 3: Latency between an initiating press and the first redundant press, pre-change against post-change; the post-change distribution shifts markedly earlier.

Across the full post-change window, the floor-throughput figure that motivated the retuning met its 46-floor-stops-per-car-per-peak-hour target in every week we logged.

## Discussion

The retuned delay achieved what it was tuned to achieve: the floor-throughput figure has sat inside its target band through the full post-change window. Redundant call-button presses, a behaviour the retuning did not target and the Living Dashboard does not track, rose sharply and then partially decayed — riders did habituate to the shorter window, but not all the way back to a rate the building had, until the retuning, treated as normal. We observe consistent improvements in most settings on the metric the retuning was designed against, provided that metric is read as a count of completed floor-stops rather than as any broader statement about how the change was received.

The floor-band result suggests the residual elevated rate is not concentrated in any one part of the building’s usage pattern — a low-floor or high-floor rider is not disproportionately responsible for it — which we take as mild evidence the effect reflects something general about the shortened confirmation window rather than a routing quirk specific to particular calls. The la-

tency compression is consistent with this reading: riders are not simply pressing more; they are also pressing sooner, as though recalibrating how long a first press should be trusted before a second one seems warranted. The partial, rather than complete, decay we observe is broadly consistent with field evidence that many behavioural interventions leave measurable residual effects well after their proximate cause is withdrawn or, as here, absorbed into a new normal [16]; our design cannot distinguish habituation to the delay itself from a more general reluctance to fully unlearn a recently useful strategy.

## Limitations

Our log covers one building’s twelve post-change weeks, and a decay model with a single exponential term may understate a more gradual, multi-timescale habituation process than our twelve weekly points can resolve, though the model’s residuals show no systematic pattern across the window studied. The one-second guard band used to define a redundant press is a judgement call; a shorter or longer guard would shift the absolute rate without, in supplementary checks at 0.5 and 2 seconds, changing the shape of the decay curve. We do not observe why any individual rider re-pressed, only that they did, and cannot rule out that some redundant presses reflect something other than a belief that the first press had not registered.

## Conclusion

Retuning a lift’s door-close delay to a floor-throughput target changed more than the figure it was aimed at: redundant call-button presses more than doubled in the week the change took effect, then declined over twelve weeks toward a rate that remains meaningfully above what the building had previously treated as baseline, with no reliable difference in that decay across floor bands. The floor-throughput figure met its target throughout. The Living Dashboard, which reports the figure the retuning was tuned against, carries no tile for the behaviour this paper logs instead. Future work might extend the decay model past twelve weeks to establish whether the residual elevation is itself stable, and might ask, directly, whether riders experienced the shorter delay as an improvement.

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